Lecture 14

Identical particles: bosons and fermions

Objectives. *Two electrons are particles of the same kind, and no experiment tells which of them is the first. This lecture finds what that does to the states of a pair. They are symmetric or antisymmetric under the swap of the two particles, and which one is a law of nature: antisymmetric for electrons, symmetric for photons. For electrons it gives Pauli's exclusion. An orbital holds at most two electrons, as Lecture 8 promised. For photons it gives an interference effect. Two photons, alike in everything but their input port, that meet at a balanced beam splitter one at each input always leave by the same output. On two sites, hopping and repulsion favour the singlet of two electrons, with no magnetic force.*

14.1 Exchange

Let two particles of the same kind each have the state space $V = \mathbb{C}^n$. By Postulate 10.1 the pair has the state space $V \otimes V$. The first factor belongs to "particle 1" and the second to "particle 2". For particles of the same kind these labels are only the slots of the tensor product. Every reading that can be made on the pair treats the two particles alike.

The *swap* W of Lecture 10, $W(u \otimes w) = w \otimes u$, exchanges the slots. It permutes the orthonormal basis $e_i \otimes e_j$, so it is unitary and self-adjoint, and $W^2 = \mathbb{1}$. Its eigenvalues are ± 1, and its two eigenspaces are the spaces of *Linear Algebra*, Def. 12.26:

$$\mathrm{Sym}^2 V = \{\psi : W\psi = \psi\}, \qquad \Lambda^2 V = \{\psi : W\psi = -\psi\}.$$

Linear Algebra writes S^2V for the first; we write $\mathrm{Sym}^2 V$, since S is a spin. For an orthonormal basis u_i of V, the vectors $u_i \otimes u_i$ and $\frac{1}{\sqrt{2}}(u_i \otimes u_j + u_j \otimes u_i)$, $i < j$, are an orthonormal basis of $\mathrm{Sym}^2 V$, and the vectors $\frac{1}{\sqrt{2}}(u_i \otimes u_j - u_j \otimes u_i)$, $i < j$, of $\Lambda^2 V$.

Two particles are *identical* if no reading tells them apart. We take this to mean that an observable A of the pair satisfies $WAW = A$, that is, $[A, W] = 0$. This is a requirement on which matrices count as readings of the pair, not a theorem. The observable $B \otimes \mathbb{1} + \mathbb{1} \otimes B$, the sum of one reading B over both particles, satisfies it, since $W(B \otimes \mathbb{1})W = \mathbb{1} \otimes B$. The observable $B \otimes \mathbb{1}$, a reading of "the first particle", does not satisfy it, unless B is a multiple of $\mathbb{1}$.

Proposition 14.1 (Exchange sectors). *Let A be an observable and H a constant Hamiltonian of the pair, both commuting with W. A reading of A that starts in $\mathrm{Sym}^2 V$ leaves the pair in $\mathrm{Sym}^2 V$, for every reading of positive probability, and the evolution $U_t = e^{-\mathrm{i}tH/\hbar}$ maps $\mathrm{Sym}^2 V$ into itself. The same holds for $\Lambda^2 V$.*

Proof. By Lemma 4.3, every projection E_a of A commutes with W. If $W\psi = \psi$, then $WE_a\psi = E_aW\psi = E_a\psi$, so Lüders' state $E_a\psi$ is symmetric. The swap W is a symmetry of H (Definition 6.3), so it commutes with every U_t, and $WU_t\psi = U_t\psi$ in the same way. For $\Lambda^2 V$, replace ψ by $-\psi$ on the right. ∎

So whether the pair is in $\mathrm{Sym}^2 V$ or in $\Lambda^2 V$ is a property that no reading and no evolution of identical particles changes. Moreover, by Proposition 6.4, the swap is a conserved observable, with the readings ± 1.

14.2 Bosons and fermions

Which sector? Experiment answers, and the answer depends only on the kind of particle.

Postulate 14.2 (Symmetrisation)**.** The states of two identical particles lie in $\mathrm{Sym}^2 V$ for some kinds of particle, called *bosons*, and in $\Lambda^2 V$ for the others, called *fermions*. Here V is the state space of one particle with all its degrees of freedom, its motion and its spin. For N identical particles the state is symmetric, for bosons, or antisymmetric, for fermions, under the swap of every two of them.

Photons are bosons. Electrons, protons and neutrons are fermions. The *spin–statistics theorem* of relativistic quantum field theory, proved by Fierz in 1939 and Pauli in 1940, says that particles of integer spin are bosons and particles of half-integer spin are fermions. We cite it. It does not follow from the finite theory of this course, and here the postulate carries it.

14.3 Pauli's exclusion

Theorem 14.3 (Pauli's exclusion)**.** *For every $u \in V$, the vector $u \otimes u$ lies in $\Lambda^2 V$ only if $u = 0$. So two identical fermions are never both in the same one-particle state.*

Proof. Since $W(u \otimes u) = u \otimes u$, a vector $u \otimes u$ in $\Lambda^2 V$ has $u \otimes u = -u \otimes u$, so $u \otimes u = 0$. By the inner product of § 10.1, its norm is $\|u\|^2$, so $u = 0$. ∎

An electron has a motion and a spin, so its state space is $V = O \otimes \mathbb{C}^2$: an *orbital* space O, for example the N sites of the ring of Lecture 8, times the spin. To use the spin states of Lecture 13, group the factors of the pair by kind. The map R from $V \otimes V = (O \otimes \mathbb{C}^2) \otimes (O \otimes \mathbb{C}^2)$ to $(O \otimes O) \otimes (\mathbb{C}^2 \otimes \mathbb{C}^2)$ with

$$R\big((o \otimes s) \otimes (o' \otimes s')\big) = (o \otimes o') \otimes (s \otimes s')$$

on products of basis vectors takes an orthonormal basis to an orthonormal basis, so it is unitary. It turns the swap of the two electrons into the swap of both pairs of factors: $RWR^\dagger = W_O \otimes W_S$, where W_O swaps the two orbitals and W_S the two spins. Since W_O and W_S commute and each has the eigenvalues ± 1, the antisymmetric states of two electrons are, after R,

$$\big(\mathrm{Sym}^2 O \otimes \Lambda^2 \mathbb{C}^2\big) \oplus \big(\Lambda^2 O \otimes \mathrm{Sym}^2 \mathbb{C}^2\big). \tag{14.1}$$

Every antisymmetric state is the sum of a vector of the first subspace and a vector of the second. In particular, a state that is a product of an orbital part and a spin part is

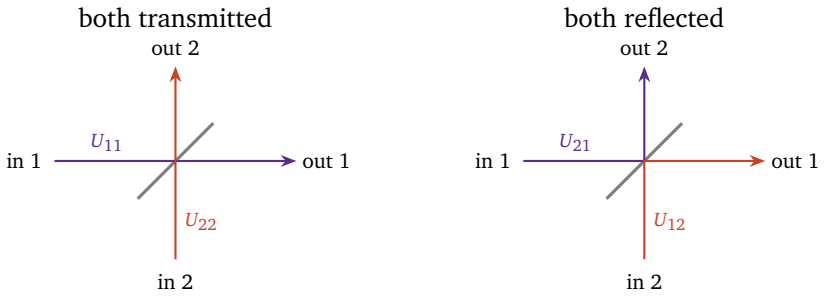


Figure 14.1: The two ways in which one particle reaches each output: both transmitted, with the amplitude $U_{11}U_{22}$, or both reflected, with $U_{21}U_{12}$. For bosons the two amplitudes add, for fermions they subtract, and for distinguishable particles their probabilities add.

antisymmetric exactly when the orbital part is symmetric and the spin part is the singlet ψ^-, or the orbital part is antisymmetric and the spin part a state of the triplet (Example 13.3).

Corollary 14.4 (Two electrons to an orbital). *Two electrons both in the orbital $u \in O$, that is, in a state $R^\dagger\big((u \otimes u) \otimes \chi\big)$, have the spin state χ on the line of the singlet ψ^-. Three electrons cannot all be in one orbital.*

Proof. The vector $u \otimes u$ is symmetric, so by (14.1) the spin part χ is antisymmetric, and $\Lambda^2\mathbb{C}^2$ is the line of ψ^-. For three electrons in the orbital u, the same grouping gives a spin part $\chi = \sum_{a,b,c} c_{abc}\,|a\rangle \otimes |b\rangle \otimes |c\rangle$ in $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$, and by Postulate 14.2 it is antisymmetric under the swap of every two spins. Swapping two slots with equal indices gives $c_{abc} = -c_{abc}$ whenever two of a, b, c are equal. The indices take only the two values 0 and 1, so in every triple two are equal, every $c_{abc} = 0$, and $\chi = 0$. ∎

So an orbital holds at most two electrons, and two electrons in one orbital have opposite spins, in the singlet. § 14.6 fills the orbitals of benzene.

14.4 Two particles at a beam splitter

A *beam splitter* is a half-silvered mirror with two input ports and two output ports. In the *ideal two-mode model* a particle's state space is $\mathbb{C}^2$, with the basis e_1, e_2: the particle is in port 1 or port 2. Every other degree of freedom, the polarisation of a photon or the shape and the arrival time of its wave packet, is the same for both particles, so its factor is one fixed vector and is left out. The splitter acts on each particle by one unitary U on $\mathbb{C}^2$, and so on the pair by $U \otimes U$. It is *balanced* if $|U_{ij}|^2 = \frac{1}{2}$ for all i, j, so that a single particle leaves by either output with probability $\frac{1}{2}$.

Two detectors, one at each output, record a *coincidence* when each fires, that is, when there is one particle at each output. Its projection is

$$E_c = E_1 \otimes E_2 + E_2 \otimes E_1, \qquad E_i = |e_i\rangle\langle e_i|\,,$$

which commutes with W. One particle enters at each input.

Theorem 14.5 (Two particles at a beam splitter). *In the ideal two-mode model, with one particle at each input, the probability of a coincidence is*

$$|U_{11}U_{22} + U_{12}U_{21}|^2 \text{ for two bosons}, \qquad |\det U|^2 = 1 \text{ for two fermions},$$

and $|U_{11}U_{22}|^2 + |U_{12}U_{21}|^2$ *for two distinguishable particles in the state* $e_1 \otimes e_2$*. For every balanced* U *these are* 0*,* 1 *and* $\frac{1}{2}$*.*

Proof. First, $Ue_j = \sum_i U_{ij} e_i$, so $(U \otimes U)(e_1 \otimes e_2) = \sum_{i,k} U_{i1}U_{k2}\, e_i \otimes e_k$. Its components on the two coincidence vectors $e_1 \otimes e_2$ and $e_2 \otimes e_1$ are $U_{11}U_{22}$ and $U_{21}U_{12}$, and those of $(U \otimes U)(e_2 \otimes e_1)$ are $U_{12}U_{21}$ and $U_{22}U_{11}$. Next, the boson state $\frac{1}{\sqrt{2}}(e_1 \otimes e_2 + e_2 \otimes e_1)$ goes to a vector with the component $\frac{1}{\sqrt{2}}(U_{11}U_{22} + U_{12}U_{21})$ on each of the two coincidence vectors. The coincidence probability is the sum of the two squared moduli, $|U_{11}U_{22} + U_{12}U_{21}|^2$. The fermion state gives, in the same way, $|U_{11}U_{22} - U_{12}U_{21}|^2 = |\det U|^2$, which is 1 for every unitary U. The state $e_1 \otimes e_2$ gives $|U_{11}U_{22}|^2 + |U_{21}U_{12}|^2$.

Finally, let U be balanced. Then $|U_{11}U_{22}| = |U_{12}U_{21}| = \frac{1}{2}$, so the distinguishable particles give $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. Two numbers of modulus $\frac{1}{2}$ whose difference has modulus $|\det U| = 1 = \frac{1}{2} + \frac{1}{2}$ are opposite, since equality holds in the triangle inequality only for opposite numbers. So $U_{12}U_{21} = -U_{11}U_{22}$, and the bosons give 0. ∎

For the balanced splitter $U = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, the two bosons leave in the state $\frac{1}{\sqrt{2}}(e_1 \otimes e_1 - e_2 \otimes e_2)$. They are both at output 1 or both at output 2, each with probability $\frac{1}{2}$, and never one at each. Figure 14.1 shows why. A coincidence happens in two ways, both particles transmitted or both reflected. For bosons the two amplitudes add, and at a balanced splitter they cancel. For fermions they subtract, and the particles always leave apart. For distinguishable particles the two ways can be told apart, and their probabilities add.

14.5 Exchange includes spin

Electrons are fermions, so one might expect two electrons at a beam splitter always to leave apart. That is true only for some spin states.

Example 14.6 (Two electrons at a beam splitter)**.** Each electron has $V = \mathbb{C}^2 \otimes \mathbb{C}^2$, port times spin, and the splitter acts on the port only, by $U \otimes \mathbb{1}$ on each electron. One electron enters at each input, and all degrees of freedom other than port and spin are in the same fixed one-particle state. We take states that are a product of a port part and a spin part. By (14.1), the port part is symmetric when the spin part is the singlet, and antisymmetric when it is a triplet state. The coincidence projection acts on the ports only.

1. *Spins in the singlet.* The port part is $\frac{1}{\sqrt{2}}(e_1 \otimes e_2 + e_2 \otimes e_1)$, and by Theorem 14.5 a balanced splitter gives coincidence probability 0, as for bosons.
2. *Both spins up.* The spin part $|\uparrow\uparrow\rangle$ is symmetric, so the port part is $\frac{1}{\sqrt{2}}(e_1 \otimes e_2 - e_2 \otimes e_1)$, and the coincidence probability is 1.

The antisymmetry is a property of the whole state, port and spin together, not of each part. Two electrons can bunch like photons, if their spins are in the singlet. The zero also needs the assumptions of the model. If the two photons differ in polarisation or arrival time, the factor left out is not the same for both, and the zero is lost (Exercise Sheet 14).

14.6 Many fermions: Slater determinants

By Postulate 14.2, the states of N identical fermions lie in $\Lambda^N V$, the vectors of $V \otimes \cdots \otimes V$ (N factors) that change sign under the swap of any two factors. For vectors $u_1, \ldots, u_N$ of V, the

normalised antisymmetrised product is

$$u_1 \wedge \cdots \wedge u_N = \frac{1}{\sqrt{N!}} \sum_\pi \operatorname{sgn}(\pi)\, u_{\pi(1)} \otimes \cdots \otimes u_{\pi(N)},$$

the sum over all orderings π of $1, \dots, N$, with the sign of π. *Linear Algebra* writes $u \wedge v$ for $u \otimes v - v \otimes u$, without the factor. For orthonormal u_i it is called a *Slater determinant*.

Proposition 14.7 (Slater determinants)**.** *The vector $u_1 \wedge \cdots \wedge u_N$ lies in $\Lambda^N V$. It is 0 if two of the u_i are equal, and a unit vector if they are orthonormal. For an orthonormal basis $u_1, \dots, u_n$ of V, the vectors $u_{i_1} \wedge \cdots \wedge u_{i_N}$ with $i_1 < \cdots < i_N$ form an orthonormal basis of $\Lambda^N V$, which has the dimension $\binom{n}{N}$ (0 for $N > n$). If $hu_i = \varepsilon_i u_i$ for a one-particle Hamiltonian h, then $H = \sum_k \mathbb{1} \otimes \cdots \otimes h \otimes \cdots \otimes \mathbb{1}$ (h in the k-th slot) has $H(u_{i_1} \wedge \cdots \wedge u_{i_N}) = (\varepsilon_{i_1} + \cdots + \varepsilon_{i_N})\, u_{i_1} \wedge \cdots \wedge u_{i_N}$.*

Proof. A swap of two slots turns the sum into the same sum with each ordering composed with a transposition, which changes every sign. If $u_i = u_j$, the orderings pair off, each with its composition with the swap of i and j, and the two terms of a pair cancel. For orthonormal u_i the $N!$ products are orthonormal, so the norm is 1. Products from different sets of indices are orthogonal. For the spanning, expand a vector of $\Lambda^N V$ in the products $u_{j_1} \otimes \cdots \otimes u_{j_N}$. A swap of two slots with equal indices shows that such coefficients are 0, and a swap of two slots multiplies a coefficient by -1, so the coefficient of any ordering of $i_1 < \cdots < i_N$ is the sign of the ordering times that of $i_1, \dots, i_N$. So the vector is a combination of the $u_{i_1} \wedge \cdots \wedge u_{i_N}$. Finally H multiplies every product of eigenvectors by the same sum of energies. ∎

So the lowest level of N fermions that do not interact fills the N lowest one-particle levels. In the Hückel model of benzene (Example 8.2) with spin, the six π electrons fill the six one-particle states $f_k \otimes |\uparrow\rangle$, $f_k \otimes |\downarrow\rangle$ with $k = 0, \pm 1$, of energies $c - 2\kappa$ and $c - \kappa$, two to each orbital. The energy is $6c - 8\kappa$, and the lowest level is a single state, since these six states hold exactly six electrons.

14.7 Two electrons on two sites

Electrons repel each other. The simplest model with repulsion has two sites a and b, the orbital space $O = \operatorname{span}(e_a, e_b)$, one electron's Hamiltonian $h = -\kappa\big(|e_a\rangle\langle e_b| + |e_b\rangle\langle e_a|\big) \otimes \mathbb{1}$ on $V = O \otimes \mathbb{C}^2$ with $\kappa > 0$, and an energy $\varepsilon_U > 0$ when both electrons are on one site:

$$H = h \otimes \mathbb{1} + \mathbb{1} \otimes h + \varepsilon_U\, (n_a \otimes n_a + n_b \otimes n_b), \qquad n_s = |e_s\rangle\langle e_s| \otimes \mathbb{1}, \tag{14.2}$$

so $n_s \otimes n_s$ projects onto "both on s". This is the Hubbard model on two sites. Hubbard writes U for ε_U, but here U is a unitary. Nothing in H acts on the spins.

Proposition 14.8 (Two electrons on two sites)**.** *On $\Lambda^2 V$, after the reordering R of (14.1):*

1. *the three states with the orbital part $\frac{1}{\sqrt{2}}(e_a \otimes e_b - e_b \otimes e_a)$ and a triplet spin part have the energy 0;*
2. *with the spin part ψ^-, the orbital parts $\frac{1}{\sqrt{2}}(e_a \otimes e_b + e_b \otimes e_a)$ (covalent) and $\frac{1}{\sqrt{2}}(e_a \otimes e_a + e_b \otimes e_b)$ (ionic) span a plane on which H has the matrix $\left(\begin{smallmatrix} 0 & -2\kappa \\ -2\kappa & \varepsilon_U \end{smallmatrix}\right)$, and $\frac{1}{\sqrt{2}}(e_a \otimes e_a - e_b \otimes e_b)$ has the energy ε_U.*

The lowest level is a single singlet state, at $\varepsilon_U/2 - \sqrt{\varepsilon_U^2/4 + 4\kappa^2}$, below the triplet by

$\sqrt{\varepsilon_U^2/4 + 4\kappa^2} - \varepsilon_U/2$.

Proof. The six states are orthonormal and lie in $\Lambda^2 V$ by (14.1), so they span it. The repulsion is 0 on orbital parts with the two electrons on different sites, and ε_U on $e_a \otimes e_a$ and $e_b \otimes e_b$. For the hops, $h \otimes \mathbb{1} + \mathbb{1} \otimes h$ takes $e_a \otimes e_b + e_b \otimes e_a$ to $-\kappa(e_b \otimes e_b + e_a \otimes e_a) - \kappa(e_a \otimes e_a + e_b \otimes e_b)$, which gives the entry -2κ. It takes $e_a \otimes e_b - e_b \otimes e_a$ to $-\kappa(e_b \otimes e_b - e_a \otimes e_a) - \kappa(e_a \otimes e_a - e_b \otimes e_b) = 0$, and $e_a \otimes e_a - e_b \otimes e_b$ to 0 in the same way. Proposition 6.11 gives the two levels of the plane. The lower is below 0, and the others are 0, ε_U and the upper level of the plane. ∎

For $\varepsilon_U \gg \kappa$ the splitting is about $4\kappa^2/\varepsilon_U$. The spins prefer to be opposite, although H has no term that acts on the spins. In the triplet the orbital part is antisymmetric, so the electrons never share a site and cannot lower their energy by hopping. This is antiferromagnetic *exchange*, from hopping, repulsion and exclusion, not from magnetic moments as is hydrogen's $\boldsymbol{\sigma} \cdot \boldsymbol{\sigma}$. With several orbitals on a site the triplet can be lower (Hund's rule). For $\varepsilon_U > 0$ the lowest state is not a single Slater determinant (Exercise Sheet 14).

The physics

Hong, Ou and Mandel sent pairs of photons from a nonlinear crystal into the two inputs of a beam splitter in 1987, and delayed one photon by a variable time. When the two wave packets overlapped in time at the splitter, the coincidences dropped nearly to zero, the dip that now carries their names. Away from the overlap the photons are told apart by their arrival times, and the coincidences return to their background level (probability $\frac{1}{2}$ in the ideal balanced model). The width of the dip, less than a picosecond, measured the length of the photons' wave packets. Pauli proposed his exclusion principle in 1925, to explain the shells of electrons in atoms, before the antisymmetric states that express it were known. In the hydrogen molecule the two electrons form a bond in the singlet, close to the covalent state of § 14.7 (Heitler and London, 1927, with orbitals that are not orthogonal). Anderson (1959) explained the antiferromagnetism of many insulators by the exchange of § 14.7.

Identical particles

States: for bosons $\mathrm{Sym}^2 V$, for fermions $\Lambda^2 V$, the eigenspaces of the swap W (Postulate 14.2); no observable A with $[A, W] = 0$ and no evolution under H with $[H, W] = 0$ leaves them (Proposition 14.1). *Pauli:* $u \otimes u \notin \Lambda^2 V$ for $u \neq 0$; an orbital holds at most two electrons, in the singlet (Theorem 14.3, Corollary 14.4). *Beam splitter:* in the ideal two-mode model, one particle at each input of a balanced splitter, coincidence probabilities 0, 1, $\frac{1}{2}$ for bosons, fermions and distinguishable particles (Theorem 14.5). *Many fermions:* Slater determinants form a basis of $\Lambda^N V$ (Proposition 14.7); on two sites the singlet is lowest (Proposition 14.8).

Summary.

- *Exchange sectors*: the symmetric and antisymmetric states are kept by every reading and evolution of identical particles (Proposition 14.1).
- *Pauli's exclusion*: no two fermions share a one-particle state; an orbital holds at most two electrons (Theorem 14.3, Corollary 14.4).

- *Two particles at a beam splitter*: in the ideal two-mode model, one particle at each input of a balanced splitter, bosons never leave apart and fermions always do (Theorem 14.5).
- *Exchange*: on two sites, hopping, repulsion and exclusion put the singlet below the triplet (Proposition 14.8).

The end of the course. Every system of this course had a finite number of levels: a qubit, n levels, a ring of N sites, pairs and triples of them, N identical particles. For these the four questions of the course, about states, observables, evolution and measurement, have answers proved from *Linear Algebra*. *Quantum Mechanics II: Continuous Systems* takes up a particle that moves on a line. Its states need a space of infinite dimension, $L^2(\mathbb{R})$. Its position observable has no eigenvectors, and its position and momentum are operators defined only on part of the space. Lecture 9 showed why the canonical relation $QP - PQ = \mathrm{i}\hbar\mathbb{1}$ needs a space of infinite dimension. The next course develops these operators and their domains.

References

Sakurai and Napolitano, *Modern Quantum Mechanics*, §7.1–7.2, pp. 429–434, §7.3, pp. 434–437, and §7.5, pp. 441–442. Cohen-Tannoudji, Diu and Laloë, *Quantum Mechanics*, vol. 2, Ch. XIV, pp. 1419–1458 (Slater determinants p. 1438). Heitler and London, *Zeitschrift für Physik* 44, 455 (1927). Anderson, *Physical Review* 115, 2 (1959). Hong, Ou and Mandel, *Physical Review Letters* 59, 2044 (1987). Pauli, *Zeitschrift für Physik* 31, 765 (1925). Pauli, *Physical Review* 58, 716 (1940).